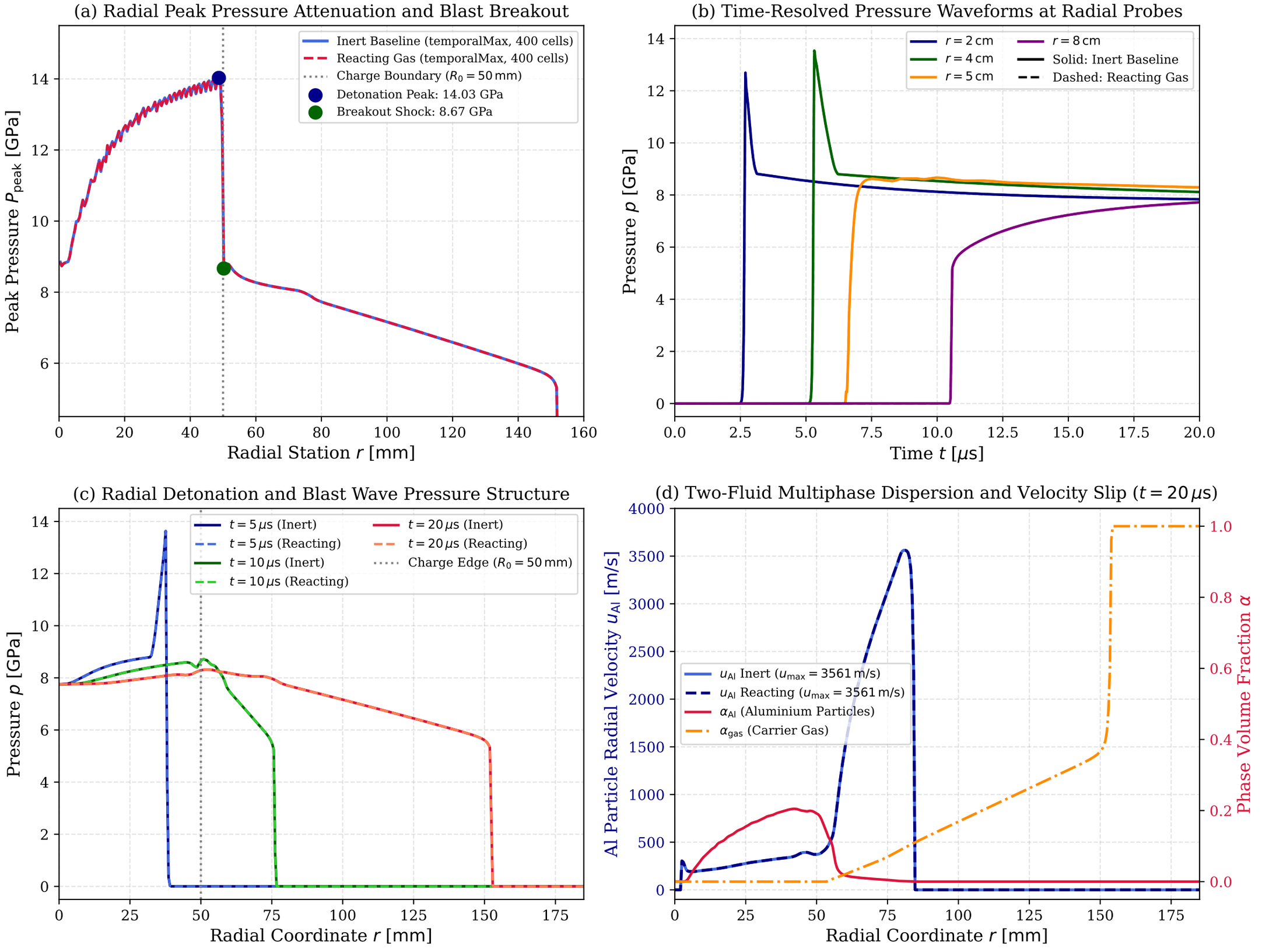
(a) Radial Peak Pressure Attenuation and Blast Breakout
Inert Baseline (temporalMax, 400 cells)
Reacting Gas (temporalMax, 400 cells)
Charge Boundary ($R_0 = 50$ mm)
Detonation Peak: 14.03 GPa
Breakout Shock: 8.67 GPa
Peak Pressure P_{peak} [GPa]
Radial Station r [mm]
(b) Time-Resolved Pressure Waveforms at Radial Probes
$r = 2$ cm
$r = 4$ cm
$r = 5$ cm
$r = 8$ cm
Solid: Inert Baseline
Dashed: Reacting Gas
Pressure p [GPa]
Time t [μs]
(c) Radial Detonation and Blast Wave Pressure Structure
$t = 5\,\mu$s (Inert)
$t = 5\,\mu$s (Reacting)
$t = 10\,\mu$s (Inert)
$t = 10\,\mu$s (Reacting)
$t = 20\,\mu$s (Inert)
$t = 20\,\mu$s (Reacting)
Charge Edge ($R_0 = 50$ mm)
Pressure p [GPa]
Radial Coordinate r [mm]
(d) Two-Fluid Multiphase Dispersion and Velocity Slip ($t = 20\,\mu$s)
u_{Al} Inert ($u_{max} = 3561$ m/s)
u_{Al} Reacting ($u_{max} = 3561$ m/s)
α_{Al} (Aluminium Particles)
α_{gas} (Carrier Gas)
Al Particle Radial Velocity u_{Al} [m/s]
Phase Volume Fraction α
Radial Coordinate r [mm]